

Progress report

sample · 2026-10-19

Your mirror, not a grade. Everything here comes from the work you chose to share; scores are the Primer's estimates.

Sample report: generated from the Primer's fixture proofs and grading cases over simulated dates (scripts/dev/primer_report_sample.py). Not a real learner.

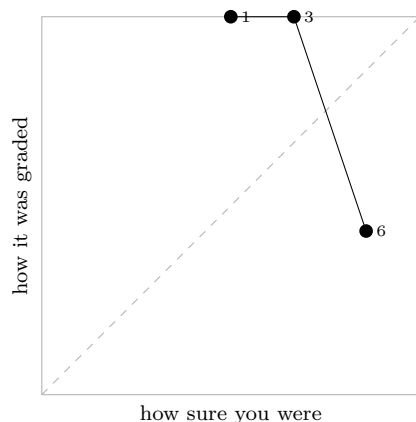
Goals and progress

- Exam: 2026-12-05, 47 days away (four 90-minute sessions of three problems).
- Your goal: Solve 3 problems cleanly; about 30/120.
- This week: 1.5 h of problems against 8 h planned.
- Problems attempted: 10; reviews scheduled: 10.

Mastery by area



Calibration



Over 10 rated proofs your average gap between confidence and grade is +10 points (positive: more sure than the grades). Brier score 0.2462 (0 is perfect).

Weak spots

- Never assumes p/q is in lowest terms (seen 2 times): $\sqrt{2}$ is irrational.
- Uses ' p^2 even $\Rightarrow p$ even' without proof (seen 2 times): $\sqrt{2}$ is irrational.

- Assumes what is to be proved (seen 1 times): $a^2 + b^2 \geq 2ab$.
- Omits or mis-states the base case (seen 1 times): Sums of odd numbers.
- Induction step does not use the induction hypothesis (seen 1 times): Sums of odd numbers.

Topics this week: Problem-solving techniques 4 attempts, 103.2 min, trend +0.01; Number theory 5 attempts, 43 min, trend +1.49; Logic and proof 1 attempts, 35 min, trend +0.23.

How you approach problems

- Reaches for proof by contradiction first 5 of 10 attempts
- Strong at proof by contradiction, parity, using definitions precisely *mastery above 0.8 over at least 3 observations each*
- Little sign of testing small cases or examples before proving *no such step in 10 attempts*
- Most marks are for missing justification rather than wrong ideas 6 of 7 marks
- Asks for hints 2 hints over 10 attempts

Things you might not see from inside

No combinatorics in the last two weeks (10 attempts elsewhere).

One short problem from it this week keeps it from going cold.

confidence 50%

Confirm or dismiss each one in the Learner tab: your verdict teaches the Primer.

Error journal

- Your error journal: Uses ' p^2 even $\Rightarrow$ p even' without proof. Where would it bite in a proof? new
- Your error journal: Never assumes p/q is in lowest terms. Where would it bite in a proof? new
- Your error journal: Assumes what is to be proved. Where would it bite in a proof? new
- Your error journal: Omits or mis-states the base case. Where would it bite in a proof? new
- Your error journal: Induction step does not use the induction hypothesis. Where would it bite in a proof? new

Next problems

1. Infinitely many primes: Prove there are infinitely many primes.

Why: Spaced review: Proof by contradiction is fading (last practised a while ago).

2. Consecutive Fibonacci numbers: Prove consecutive Fibonacci numbers are coprime.

Why: Spaced review: Mathematical induction is fading (last practised a while ago).

3. $\sqrt{3}$ is irrational: Prove that $\sqrt{3}$ is irrational.

Why: Targets a weak spot: Irrationality proofs (mastery 0.61).

4. A recursive sequence: Let $a_1 = 1$ and $a_{n+1} = \sqrt{2 + a_n}$. Prove the sequence converges and find its limit.

Why: Targets a weak spot: Mathematical induction (mastery 1.00).

5. Fixed points: Find the expected number of fixed points of a uniformly random permutation of $\{1, \dots, n\}$.

Why: At the edge of what you can do: about a 38% chance, stretches Linearity of expectation.

Your reflection

Answer by hand. The Primer reads these only if you ask it to.

What went well in this session?

Where did you get stuck, and what got you unstuck?

What will you do differently next time?
